frontiers in Education

TYPE Review
PUBLISHED 24 June 2026
DOI 10.3389/feduc.2026.1823241

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OPEN ACCESS

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RECEIVED 04 March 2026
REVISED 05 June 2026
ACCEPTED 05 June 2026
PUBLISHED 24 June 2026

CITATION
Massalimova B, Darmenbayeva A, Afanasenkova I, Dautova Z, Zhurkhabaeva L, Zhanaliyeva R, Imangaliyeva B, Shaikhova B, Ungarbayeva A, Kabylbekova A, Almuratova K and Askarova M (2026) Integration of STEM technologies in chemistry education: opportunities, challenges, and future directions. Front. Educ. 11:1823241. doi: 10.3389/feduc.2026.1823241

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Integration of STEM technologies in chemistry education: opportunities, challenges, and future directions

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The integration of STEM technologies into chemistry education is increasingly viewed as a powerful driver of transformation, capable of enhancing traditional teaching and supporting the development of critical thinking, innovative problem-solving, and practical skills. This review focuses on the pedagogical integration of STEM-related technologies in chemistry education, particularly virtual laboratories, simulations, molecular visualization systems, and mathematical modeling tools, and analyzes their role in supporting conceptual understanding, inquiry-based learning, and interdisciplinary competencies. A narrative review approach was applied to analyze recent studies on the implementation of virtual laboratories, digital simulations, molecular visualization tools, and mathematical modeling platforms in chemistry teaching, with attention to their pedagogical functions and learning outcomes. The synthesis of evidence demonstrates that STEM-based technologies improve accessibility, support practical and inquiry-based learning, enhance teamwork, and facilitate the understanding of abstract chemical concepts through interactive and visual tools. However, several challenges remain, including a shortage of qualified teachers, high financial costs, uneven technological infrastructure, and students' difficulties adapting to technology-rich environments. The review concludes that effective integration of STEM technologies in chemistry education requires targeted professional development, strategic investment in infrastructure, curriculum refinement, and closer collaboration between educational institutions and industry.

KEYWORDS
chemistry education, digital laboratories, educational challenges, inquiry-based learning, STEM technologies

1 Introduction

Modern education is undergoing a profound transformation driven by rapid technological development, global digitalization, and the growing demand for interdisciplinary competencies. As societies face increasingly complex scientific, environmental, and technological challenges, educational systems are required to prepare learners not only with disciplinary knowledge but also with the ability to integrate concepts across fields. In this context, STEM education (Science, Technology, Engineering, Mathematics) has emerged as a central paradigm for rethinking how science is taught and learned in contemporary schools and universities (Bybee, 2013; Xie et al., 2015). STEM-based approaches promote inquiry, creativity, critical thinking, and problem-oriented learning, allowing students to engage with authentic scientific problems through interdisciplinary lenses (Martín-Páez et al., 2019; Tytler, 2020).

Within the broader STEM framework, chemistry holds a unique position as a discipline that bridges fundamental scientific theories with real-world technological and industrial applications. Researchers increasingly emphasize that chemistry is not merely one component of STEM but a central science that naturally intersects with engineering, environmental sciences, material science, and technological innovation (Wilson et al., 2014). This makes it particularly suitable for the integration of modern STEM technologies, which can significantly enhance the conceptual understanding of molecular processes, support safe and effective experimentation, and strengthen connections between theory and practice (Fitriyana et al., 2024; Космодемьянская and Емельянова, 2025).

Recent literature highlights significant advances in the use of digital tools in chemistry education, including virtual laboratories, interactive simulations, computer-based modeling, and molecular visualization systems (Acevedo et al., 2024; Alhashem and Alfailakawi, 2023; Chan et al., 2021; Hu-Au and Okita, 2021). These technologies address long-standing challenges in chemistry teaching – such as the abstract nature of chemical concepts, high laboratory costs, and safety constraints – by providing accessible and pedagogically rich learning environments. In addition, emerging tools such as robotics, 3D printing, and data-driven modeling further expand the possibilities for integrating authentic STEM practices into chemistry education, supporting the development of interdisciplinary and technological competencies. These tools are particularly important in chemistry education, where understanding depends on the integration of macroscopic observations, submicroscopic processes, and symbolic representations (Gilbert and Treagust, 2009; Johnstone, 1991).

In this review, the term "STEM technologies" refers specifically to digital and computational tools that support the implementation of STEM-oriented pedagogy in chemistry education. Based on prior literature on technology-enhanced chemistry education and refined through the thematic analysis conducted in this review, the main categories of STEM-related technologies include virtual laboratories, simulation environments, molecular visualization systems, and mathematical modeling tools (Acevedo et al., 2024; Alhashem and Alfailakawi, 2023; Chan et al., 2021). It is important to distinguish between STEM as a pedagogical framework and the technological tools used to support it. While STEM education emphasizes interdisciplinary problem-solving, the focus of this review is on how specific technologies facilitate chemistry learning within this framework, particularly in supporting students' understanding of complex chemical phenomena across different levels of representation. Accordingly, this review focuses specifically on the pedagogical role of STEM-related technologies in chemistry education, rather than on STEM education as a general framework or on technological tools in isolation.

However, despite the promising potential of these innovations, several studies point to persistent barriers that complicate the effective integration of STEM technologies into chemistry instruction. These include limited technological infrastructure, insufficient funding, lack of teacher preparation, curriculum misalignment, and student difficulties in adapting to technology-enhanced learning environments. Researchers also note that many existing implementations remain fragmented or experimental, highlighting the need for systematic, theoretically grounded, and pedagogically meaningful approaches to STEM integration.

Despite the growing number of publications on STEM-based learning, existing review studies often focus either on general STEM education frameworks or on isolated digital tools without specifically analyzing their pedagogical integration in chemistry education. In particular, previous reviews rarely address how STEM technologies support students' understanding of chemistry-specific representational levels, namely the connections between macroscopic observations, submicroscopic molecular processes, and symbolic representations. Furthermore, limited attention has been given to the pedagogical integration of multiple STEM components, such as virtual laboratories, molecular visualization, and mathematical modeling, within a unified chemistry learning framework. For example, Martín-Páez et al. (2019) provided a conceptual overview of STEM education, while Chan et al. (2021) focused primarily on virtual laboratories as separate technological tools. Similarly, Li et al. (2020) analyzed trends in STEM education research, but did not examine discipline-specific methodological implications for chemistry learning. More recent systematic reviews have investigated immersive technologies such as virtual reality or digital simulations in STEM education (Acevedo et al., 2024), yet these studies typically emphasize technological effectiveness rather than pedagogical synthesis across multiple STEM components. As a result, the literature lacks an integrated analytical framework that connects digital laboratories, mathematical modeling, engineering design approaches, and interdisciplinary pedagogy specifically within chemistry education.

Therefore, this review aims to bridge this gap by providing a structured synthesis of recent empirical and conceptual studies on STEM technology integration in chemistry teaching. Unlike prior reviews that examine individual tools or general STEM frameworks, this study develops an analytical perspective that links technological tools with pedagogical functions and learning outcomes in chemistry education. Particular attention is given to how virtual laboratories, simulation environments, and mathematical modeling tools support conceptual understanding of abstract chemical phenomena and the development of interdisciplinary competencies.

The review is guided by the following research questions:

RQ1. What categories of STEM technologies are applied in chemistry education?

RQ2. How do STEM-related technologies support students' conceptual understanding and inquiry-oriented learning within chemistry education?

RQ3. What key challenges, limitations, and future research directions are identified in the literature on STEM technology integration in chemistry education?

2 Methods

This review was conducted using a narrative review methodology, which is appropriate for synthesizing interdisciplinary educational research characterized by diverse study designs, heterogeneous pedagogical contexts, and varied technological tools (Ferrari, 2015; Snyder, 2019). The choice of a narrative approach is motivated by the complexity and interdisciplinary nature of research devoted to the integration of STEM technologies into chemistry education, which includes empirical studies, conceptual works, methodological developments, and research focused on educational technologies. Due to the diversity of methodological approaches and differences in research designs, conducting a rigorous meta-analysis seems limited in its applicability, whereas narrative synthesis allows us to identify general pedagogical patterns, conceptual connections, and promising directions for further research.

The review covers research on the application of STEM technologies in both secondary and higher education, as modern digital tools, virtual laboratories, modeling methods, and interdisciplinary educational approaches are being actively implemented at various levels of chemistry education. This approach allows for a more holistic understanding of how STEM-related technologies support conceptual understanding, inquiry-oriented learning, and the implementation of interdisciplinary teaching approaches in chemistry education.

A search for scientific publications was conducted in the international databases Web of Science, Scopus, and Google Scholar for the period 2013–2025, which corresponds to the most active stage of development of digital educational technologies, virtual laboratories, and interdisciplinary STEM approaches in science education. The initial database search identified a total of 428 records. After removal of duplicates ($n = 76$), 352 records remained for title and abstract screening. Following the screening process, 127 articles were selected for full-text assessment. Of these, 68 studies met the inclusion criteria and were included in the final analysis. The search strategy included combinations of keywords reflecting various aspects of the integration of STEM technologies into chemistry education, including: "STEM education," "chemistry education," "technology-enhanced learning," "virtual laboratory," "digital laboratory," "simulation-based learning," "molecular visualization," "mathematical modeling," and "interdisciplinary science teaching." To ensure comprehensive coverage of scientific publications, the logical operators AND and OR were used to take into account the interdisciplinary nature of the subject matter. To ensure a focused analysis, the search strategy combined general terms (e.g., "STEM education") with discipline-specific keywords such as "chemistry education," "virtual chemistry laboratories," and "chemical modeling in education."

The study selection process followed a structured screening procedure analogous to PRISMA principles, including identification, screening, eligibility assessment, and final inclusion (Page et al., 2021). Although this review adopts a narrative approach, efforts were made to ensure transparency and reproducibility of the selection process.

Inclusion criteria included peer-reviewed scientific publications devoted to chemistry education in the STEM context, as well as studies examining the use of digital technologies, modeling, virtual laboratories, and interdisciplinary educational strategies to foster a conceptual understanding of chemical processes. The analysis included both empirical studies and theoretical and review papers that were relevant to understanding the pedagogical mechanisms of STEM technology integration. Publications in English published between 2013 and 2025 were considered.

Exclusion criteria included works not related to an educational context, studies devoted exclusively to chemistry experiments without pedagogical interpretation, and publications that did not contain a sufficient methodological description. Individual conceptual sources were included in the analysis if they represented widely cited theoretical models of STEM education or revealed fundamental approaches to interdisciplinary learning.

The study selection process is presented in Figure 1.

The publication selection procedure included a sequential analysis of titles, abstracts, and full texts to determine their relevance to the study objectives. The first stage involved screening titles to identify publications devoted to STEM approaches or the use of educational technologies in chemistry teaching. In the second stage, abstracts were analyzed to determine the pedagogical focus of the study. Finally, the full texts of the publications were assessed for relevance to the topic of integrating STEM technologies into chemistry education.

Thematic analysis was conducted to systematically examine the reviewed studies and identify major themes relevant to the research questions (Braun and Clarke, 2006). The analysis involved several stages, including familiarization with the data, generation of preliminary codes, development of thematic categories, and refinement of emerging themes. Coding was performed inductively, allowing categories to emerge from the data rather than being predefined. The main coding categories included:

(i) types of STEM technologies,
(ii) pedagogical functions,
(iii) learning outcomes,
(iv) identified challenges and limitations.

These categories were iteratively refined to ensure consistency and alignment with the research questions. This process enabled a structured and transparent synthesis of heterogeneous studies. Thematic analysis allowed us to systematize current research trends, identify the main advantages and challenges of integrating STEM technologies into chemistry education, and identify promising areas for further research.

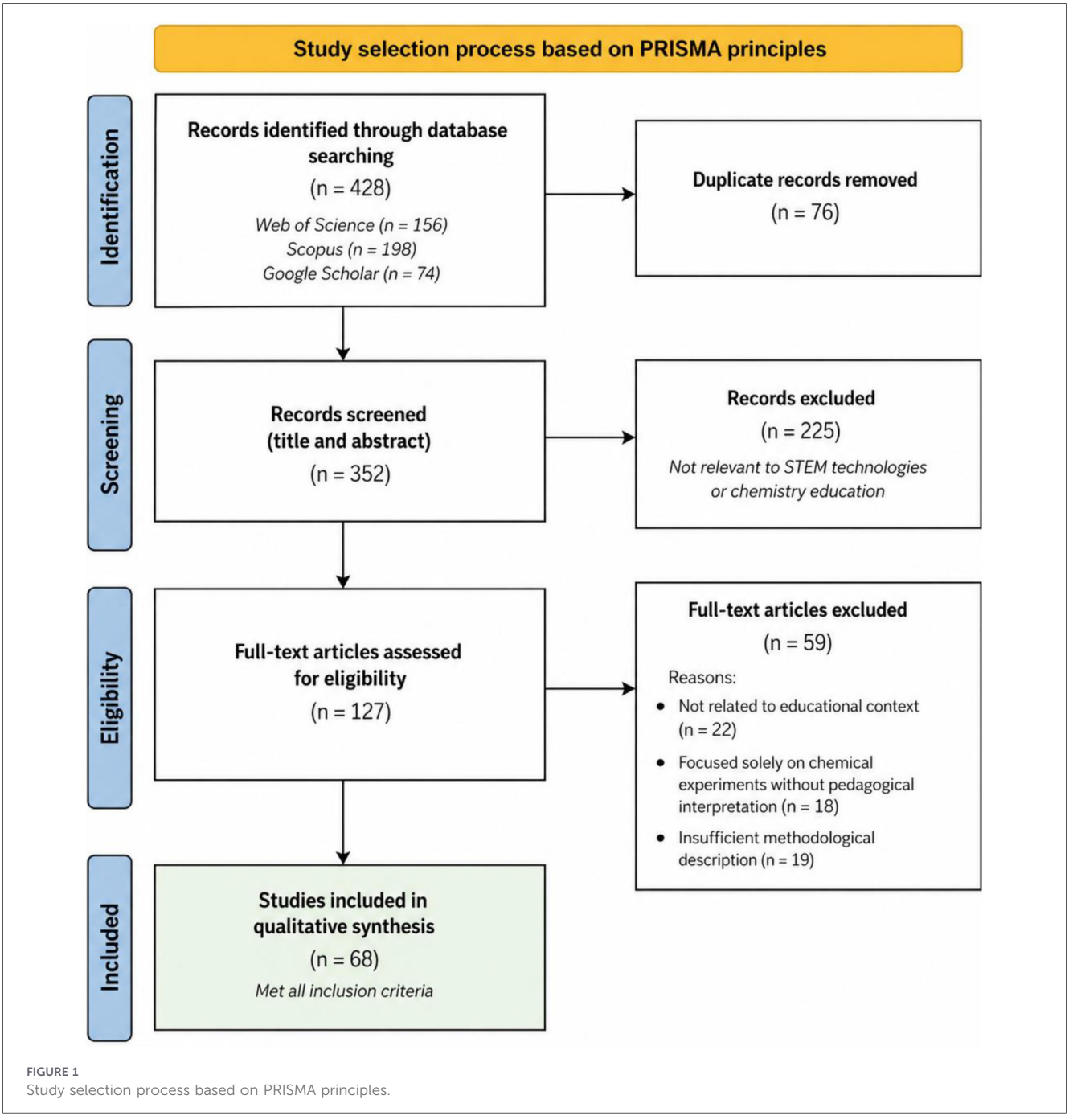


FIGURE 1
Study selection process based on PRISMA principles.

The review includes studies from both secondary and higher education contexts in order to capture a broader perspective on the implementation of STEM technologies in chemistry learning.

To enhance the rigor of the review, a qualitative appraisal of the included studies was conducted. Studies were evaluated based on relevance to chemistry education, clarity of methodological design, and alignment with the research questions. Particular attention was given to studies published in leading chemistry education journals, including Chemistry Education Research and Practice and the Journal of Chemical Education. Rather than applying a formal scoring system, studies were critically examined to identify strengths, limitations, and methodological consistency across the literature. This approach is consistent with narrative review methodology, where emphasis is placed on interpretive synthesis rather than quantitative aggregation.

3 Fundamentals of STEM education

STEM education is widely recognized as an interdisciplinary educational approach that integrates science, technology, engineering, and mathematics to promote problem-oriented learning and the development of analytical competencies (Kelley and Knowles, 2016; Li et al., 2020). However, within the broader STEM framework, chemistry occupies a distinctive position as a

central science that connects physical, biological, environmental, and materials-related processes. Unlike many other STEM disciplines, chemistry requires learners to interpret phenomena across multiple levels of representation, including observable experimental processes, symbolic language, and submicroscopic molecular structures (Figure 2). This multi-level nature of chemical knowledge creates significant cognitive challenges for learners, particularly at the submicroscopic level, where chemical processes and molecular interactions cannot be directly observed. Students often experience difficulties in connecting macroscopic observations with symbolic representations and underlying molecular-level explanations. Therefore, chemistry education requires pedagogical approaches that support visualization, modeling, and conceptual integration across these representational levels (Akrami, 2022; Gilbert and Treagust, 2009; Sutaphan and Yuenyong, 2019; Taber, 2013).

Chemistry learning involves interpretation of phenomena at macroscopic, submicroscopic, and symbolic levels. The relationships between macroscopic, submicroscopic, and symbolic representations are dynamic and interconnected rather than linear. STEM technologies such as molecular visualization systems, simulations, and virtual laboratories support students in moving between these representational levels by making abstract molecular processes more accessible and visually interpretable.

The epistemological structure of chemistry differs from that of many other STEM disciplines because chemical knowledge is constructed through the interaction between observable phenomena and theoretical models that cannot be directly perceived. Experimental results provide macroscopic evidence of chemical processes, while molecular-level interpretations explain reaction mechanisms, structural transformations, and intermolecular interactions. Symbolic representations, including equations, graphs, and mathematical relationships, allow scientists to formalize chemical knowledge and predict system behavior. Effective chemistry learning therefore requires coordinated transitions between these representational levels, which can be difficult for students due to the abstract nature of molecular concepts and the need to integrate qualitative and quantitative reasoning.

From a STEM perspective, chemistry learning involves not only understanding theoretical principles but also applying technological and mathematical tools to analyze, visualize, and predict chemical behavior. Mathematical relationships are essential for describing reaction kinetics, equilibrium systems, and thermodynamic dependencies, while engineering approaches enable the practical application of chemical knowledge to real-world challenges such as materials design, environmental protection, and energy technologies (Ambrož et al., 2023; Pernaa, 2022). The interdisciplinary character of chemistry makes it particularly suitable for integration with computational modeling, digital experimentation, and data-driven analysis.

Technological tools play a crucial role in supporting understanding of abstract chemical concepts by enabling visualization of molecular structures, simulation of reaction pathways, and representation of dynamic processes that cannot be directly observed. Digital laboratories and simulation environments allow learners to explore complex chemical systems in a safe and accessible format, facilitating connections between theoretical models and experimental observations (Habiddin and Nagol, 2023). Such tools can reduce cognitive load associated with symbolic interpretation and support inquiry-based learning by enabling students to test hypotheses and analyze relationships between variables.

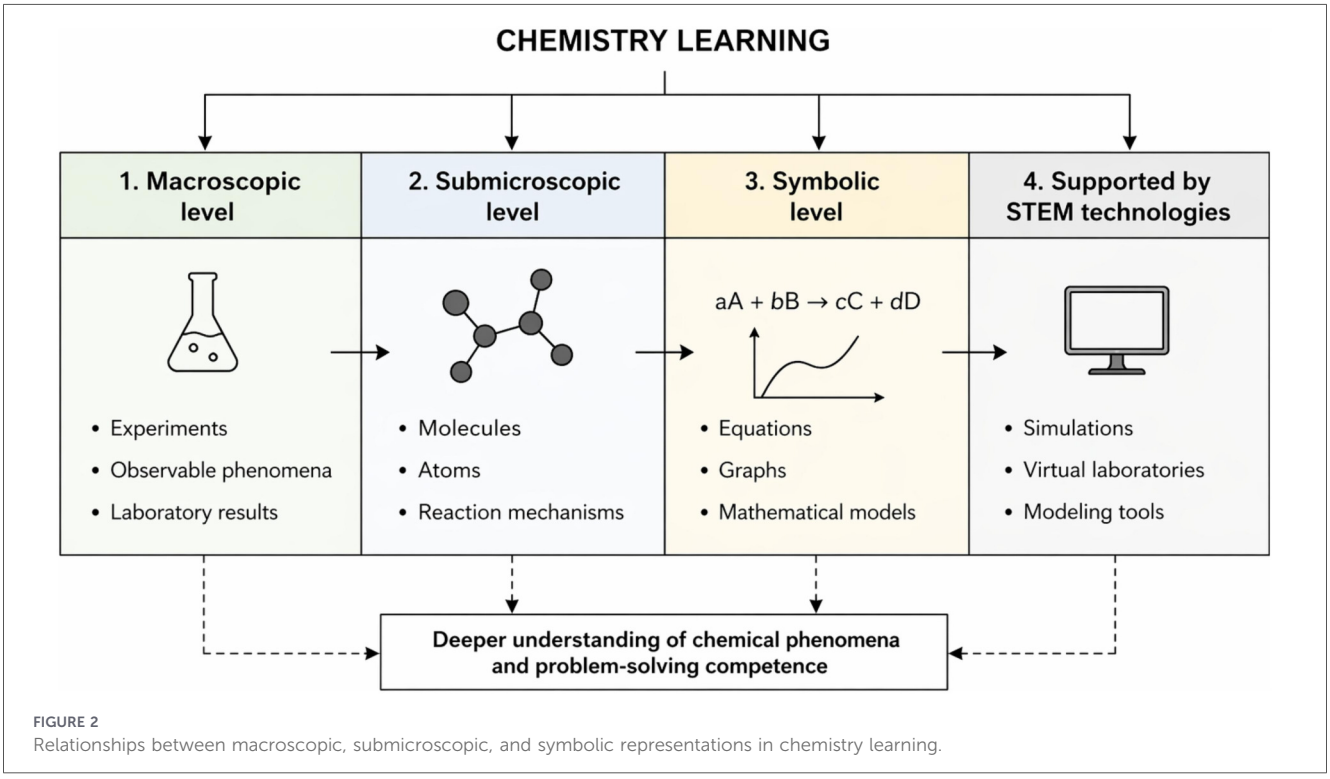


FIGURE 2
Relationships between macroscopic, submicroscopic, and symbolic representations in chemistry learning.

Understanding the specific epistemological characteristics of chemistry within STEM education provides a conceptual basis for evaluating the pedagogical role of digital technologies. The integration of simulation tools, molecular visualization systems, and mathematical modeling environments allows learners to connect macroscopic observations with molecular-level explanations and symbolic representations. This perspective enables a more precise analysis of how STEM technologies contribute to conceptual understanding, interdisciplinary competence development, and problem-oriented learning in chemistry education.

4 Application of STEM technologies in teaching chemistry

This section addresses RQ1 and RQ2 by examining the main categories of STEM-related technologies used in chemistry education and their pedagogical roles in supporting conceptual understanding and inquiry-oriented learning.

The analysis focuses on virtual laboratories, simulation environments, molecular visualization systems, and mathematical modeling tools identified through the thematic synthesis of the reviewed studies. The introduction of STEM technologies into chemistry teaching significantly expands the learning experience, allowing students to not only master theoretical concepts but also develop practical skills through the use of advanced methods such as simulations (Fabeku and Enyeasi, 2024), virtual laboratories (Rahmadani et al., 2021), 3D printing (Pinger et al., 2020), and integration with various sensors. These technologies make learning more interactive, accessible, and dynamic, offering students unique tools to explore chemical processes in real time and with high accuracy. Across recent studies, there is a growing consensus that the effectiveness of such technologies depends not merely on their availability, but on their pedagogical integration into the learning process. In particular, evidence from leading chemistry education journals, including "Chemistry Education Research and Practice" and the "Journal of Chemical Education", highlights the importance of visualization tools and modeling approaches in supporting students' understanding of abstract chemical concepts and multilevel representations.

4.1 Scientific approach and engineering principles in chemistry teaching

Chemistry as a science is based on a rigorous scientific approach, which includes hypotheses, experiments, observations, data analysis, and drawing conclusions. In chemistry education, it is important not only to master theoretical knowledge, but also to be able to apply scientific methods to solve real-world problems. The scientific approach to chemistry is the basis for understanding chemical processes (Reid, 2008). It begins with formulating a hypothesis about how a certain process might occur and then moves through the stages of conducting experiments, analyzing the results, and drawing conclusions. It is a continuous cycle that allows for a deeper understanding of molecules, reactions, and their behavior under different conditions. A critical part of the scientific method is the ability to repeat experiments to confirm or disprove theories. However, recent educational research suggests that in modern STEM contexts, this traditional cycle is increasingly mediated by digital tools that allow students to explore chemical processes beyond the constraints of physical laboratories.

In the article (Abungu et al., 2014), the Scientific Approach was applied to chemistry teaching through the Science Process Skills Teaching Approach (SPSTA). This method focused on engaging students in hands-on activities related to Volumetric and Qualitative analysis, which are part of the Kenya Certificate of Secondary Education (KCSE) curriculum. By incorporating inquiry-based learning, students were encouraged to formulate questions, conduct experiments, and analyze data, fostering critical thinking and scientific reasoning. The results of implementing SPSTA were significant; students who participated in this approach demonstrated higher achievement levels in Chemistry compared to those who were taught using traditional methods. Statistical analyses revealed that students in the experimental groups scored significantly better on the Chemistry Achievement Test, indicating that active engagement through practical activities contributed to improved understanding and retention of chemistry concepts. Furthermore, the findings suggested that this approach not only elevated academic performance but also increased students' interest in pursuing science subjects. The study emphasized that employing the Scientific Approach can effectively address challenges in student performance in chemistry and enhance the overall learning experience.

Across the reviewed studies, a consistent pattern can be identified: inquiry-based and scientific approaches significantly improve students' understanding of chemistry when learners are actively engaged in experimentation and data analysis. Despite differences in instructional design, these approaches converge in promoting higher-order thinking skills, including hypothesis formulation, critical analysis, and problem solving. At the same time, the effectiveness of such approaches varies depending on the level of guidance provided, suggesting that structured support and scaffolding are essential for achieving meaningful learning outcomes.

Nevertheless, several limitations are also evident. While most studies report positive learning outcomes, they often rely on controlled experimental conditions and may not fully reflect real classroom constraints. In addition, the successful implementation of scientific approaches requires significant time, resources, and teacher expertise, which may limit their scalability in practice.

Importantly, recent evidence indicates that STEM technologies can enhance the effectiveness of scientific approaches by enabling students to manipulate variables, visualize abstract processes, and test hypotheses in interactive environments. However, without appropriate pedagogical design, the use of technology may remain superficial and not lead to deeper conceptual understanding.

Fitriyana et al. discussed the use of an integrated STEM (science, technology, engineering, mathematics) approach in chemistry education, highlighting its potential to improve teaching and learning (Wibawa and Kardipah, 2018). The researchers employed an explorative survey method, utilizing a

Perception Scale on STEM (PC-STEM) with both closed and open-ended questions to gather insights from in-service chemistry teachers. Data analysis included one-way MANOVA testing and descriptive statistics, revealing patterns in teachers' perceptions. Teachers were encouraged to share practical examples, such as creating antioxidant ice cream, demonstrating hands-on, inquiry-based learning. The findings indicated that both high school chemistry teachers (HSCT) and vocational school chemistry teachers (VSCT) held positive views on integrated STEM learning, suggesting benefits for chemistry instruction. However, many teachers reported limited experience in applying STEM, with only 46% of HSCT and 31% of VSCT having implemented it in practice. Importantly, there were no significant differences in perceptions between HSCT and VSCT, indicating shared challenges. The study highlighted the necessity for ongoing professional development to enhance teachers' competencies in STEM integration, bridging the gap between theory and practice in chemistry education. However, these findings also reveal an important limitation. Although teachers generally recognize the value of STEM integration, actual implementation remains relatively low. This suggests a gap between theoretical acceptance and classroom practice, which may be associated with insufficient training, limited resources, or lack of confidence in applying interdisciplinary approaches.

In addition, the use of a scientific approach in chemical teaching develops students' analytical thinking, ability to independently solve problems and conduct scientific research (Clucas and Sjöström, 2024). Students learn not only to perceive information, but also to critically evaluate it, build hypotheses, and then test them in practice, which is the basis of scientific activity.

The article illustrates the effective use of a scientific approach in teaching chemistry through project-based learning (PBL) focused on distillation (Situmorang et al., 2022). This method engaged students actively in their learning by providing structured project packages that included guidelines, assessment rubrics, and evaluation tests, mirroring the scientific method of hypothesis testing and experimentation. Students worked both collaboratively and independently, developing crucial technical, critical thinking, and problem-solving skills. They collected and analyzed data from their projects, which deepened their understanding of the subject and enhanced their scientific inquiry abilities. The results showed impressive improvements, with critical thinking skills averaging a score of 84.51 ± 7.41. Additionally, students received very good classifications for their project proposal portfolios and reports, with scores of $M = 88.46 \pm 6.74$ and $M = 92.85 \pm 1.98$, respectively. These outcomes highlight the success of integrating a scientific approach in teaching, significantly boosting students' competence in analytical chemistry and preparing them for future careers in the field.

In the article (Ibrahim and Mohd Badli, 2025), the authors employed a blended problem-based learning (BPBL) module integrated with the Socratic method to enhance chemistry education. This approach aimed to connect abstract chemical concepts to real-world scenarios, thereby fostering students' problem-solving skills. By encouraging dialogue and critical questioning, students engaged actively in discussions, which improved their understanding of the microscopic, macroscopic, and symbolic representations of chemistry. The results demonstrated a significant enhancement in student engagement and comprehension. The Socratic method enabled students to articulate detailed justifications for their answers, overcoming previous challenges where they struggled to provide thorough explanations. Additionally, students were better able to connect different levels of chemical representation, leading to improved problem-solving abilities in STEM contexts. This innovative approach effectively transformed the learning environment, making chemistry more accessible and engaging for students.

Problem solving is an integral part of the scientific approach and an important component in the context of STEM education (Raman et al., 2024; Tan et al., 2023). This process begins with identifying and formulating a problem, which can be both theoretical and practical. The scientific method is based on consistent analysis and search for solutions to specific problems, which promotes a deep understanding of processes and allows students to learn to apply their knowledge to solving real-world issues.

In chemistry education, problem solving begins with identifying a problem, which can be related to the study of chemical reactions, the synthesis of new materials, the improvement of existing technologies, or the solution of environmental problems. For example, the task of finding effective methods for cleaning water from pollutants requires not only knowledge of chemistry, but also the application of approaches from other disciplines, such as engineering, ecology, and biology (Nehru et al., 2024). After formulating the problem, the hypothesis generation stage follows, when students, relying on their knowledge, put forward possible ways to solve the problem. At this stage, it is important to consider not only theoretical data, but also the practical applicability of the proposed solutions. In chemistry, this could be, for example, the choice of a type of catalyst to speed up a reaction or the use of new materials for filtration (Tantayanon et al., 2024).

Then comes the stage of planning an experiment or developing a model with which to test the hypotheses. Students begin to design an experiment in which various parameters such as temperature, pressure, and concentration of substances will be varied in order to study the influence of these factors on the result. At this stage, it is important to use not only theoretical knowledge, but also the ability to build experimental setups that will allow the study to be conducted most effectively. After conducting the experiment, the stage of data analysis and drawing conclusions begins. At this stage, students interpret the results obtained, compare them with the original hypotheses and theoretical predictions. If the experiment confirmed the hypothesis, they can offer recommendations on how to apply the knowledge gained in practice. If the results do not match the hypothesis, students must return to the hypotheses, revise the research methodology, and repeat the process.

Problem solving as part of the scientific approach in STEM education provides many advantages, such as developing critical thinking, innovation and interdisciplinary skills, as well as preparing students for real-world problems. However, this approach also has its drawbacks, including high time and resource requirements, as well as difficulties in evaluating and standardizing solutions. Despite this, the use of problem solving in teaching contributes to a deeper understanding of the

material and the development of practical skills, which makes this approach important and necessary in STEM education. Importantly, recent studies indicate that the effectiveness of scientific and engineering approaches can be significantly enhanced through the use of STEM technologies. Digital tools such as simulations, virtual laboratories, and data analysis platforms allow students to manipulate variables, test hypotheses, and visualize abstract chemical processes in ways that are not always feasible in traditional laboratory settings. However, the literature also suggests that technology integration alone does not guarantee improved learning outcomes. Without appropriate pedagogical design and guidance, students may engage with technological tools superficially, limiting their conceptual understanding. Therefore, the success of STEM-based chemistry instruction depends on the alignment between technological tools and instructional strategies.

4.2 Virtual labs in chemistry education

This category of STEM-related technologies primarily addresses RQ1 by representing one of the most frequently reported technological approaches in chemistry education. At the same time, the reviewed studies indicate that virtual laboratories support RQ2 by facilitating inquiry-oriented learning, conceptual visualization, and interactive experimentation.

Virtual laboratories are becoming an integral part of modern chemistry education, offering students unique opportunities to conduct chemistry experiments in a safe and controlled environment. This technology is actively used to overcome the limitations of traditional laboratories, such as cost, equipment availability, safety, and environmental risks. Virtual laboratories provide students with tools for deep understanding of chemical processes, experimental methods, and creation of scientific models without the need for physical presence in the laboratory (Chan et al., 2021; Hu-Au and Okita, 2021).

A virtual laboratory is a digital environment where students can interact with chemicals and reagents, conduct reactions, observe their results, and analyze data obtained during experiments (Alhashem and Alfailakawi, 2023). These labs can simulate real chemical processes such as titration, neutralization reaction, extraction, precipitation, and many others (Serrano-Perez et al., 2023). Virtual labs allow to change the parameters of the experiment (temperature, pressure, concentration), which helps students understand how these changes affect the outcome of the reaction (Rahmawati et al., 2022).

Virtual laboratories serve as an effective educational tool that enhances academic learning by addressing certain limitations found in traditional laboratory settings. When used properly, these labs provide opportunities for hands-on practice in a more cost-effective and secure environment, especially when they are made available at a reasonable price (Tatli and Ayas, 2010). Although virtual labs significantly support and improve science education through simulations, they are not intended to replace conventional laboratories entirely (Ratamun and Osman, 2018). From a chemistry education perspective, the pedagogical value of virtual laboratories lies not only in accessibility and safety but also in their ability to support students' understanding of relationships between macroscopic observations, symbolic representations, and submicroscopic molecular processes. By visualizing otherwise invisible molecular interactions and reaction mechanisms, virtual laboratories can help reduce conceptual difficulties associated with abstract chemical phenomena and support transitions between different representational levels of chemical knowledge.

Recent advances in virtual reality (VR) technologies further extend the educational potential of virtual laboratories by improving user interaction fidelity and immersion. Modern VR chemistry environments increasingly incorporate natural interaction techniques, including hand tracking, gesture-based manipulation, haptic feedback, and spatial navigation, allowing students to interact with virtual laboratory equipment in ways that closely resemble real-world experimentation. For example, recent developments such as ChemistryVR have demonstrated how immersive VR environments can support active learning, procedural understanding, and engagement through realistic laboratory simulations and intuitive interaction paradigms (Jensen and Konradsen, 2018; Jiang et al., 2024; Radianti et al., 2020). These advances suggest that next-generation virtual laboratories may provide not only conceptual visualization but also richer experiential learning opportunities that bridge the gap between virtual and physical laboratory practice.

Nevertheless, computer simulations or laboratories help overcome several challenges encountered in traditional lab work and positively support the attainment of educational goals. They also reduce mistakes caused by unfavorable lab conditions or improper use of equipment (Kolil and Achuthan, 2024). A major advantage of virtual labs is their ability to safely simulate hazardous experiments, allowing students to design, conduct, and analyze them digitally. Additionally, virtual environments make it possible to carry out complex experiments more efficiently, saving time, reducing costs, and boosting student engagement – especially when real labs suffer from equipment shortages, high expenses, or safety risks (Akinola et al., 2020). Figure 3 shows a comparison diagram of a traditional and virtual chemistry laboratory, the teacher's role, and the students' actions during the experiment.

A traditional chemistry lab allows students to work with real chemical reagents and equipment. For example, during the titration of an acid and base, students can practice precise dosing techniques and learn how to work with various lab tools. Such experiences help develop practical skills and acquire knowledge that can be useful in the future (Manyilizu, 2023). However, despite all the advantages, a traditional lab has its limitations. Firstly, there is a possibility of measurement errors, for example, due to improper dosing, which can lead to inaccurate results. Secondly, traditional labs require the use of chemical reagents, which are often expensive and can be hazardous, which creates safety risks. In addition, the waste generated after experiments requires safe disposal, which creates additional problems for the lab (Faulconer et al., 2018). Also, such experiments are time-consuming, especially if students need to repeat the titration to obtain accurate results.

On the other hand, a virtual lab offers many advantages, especially in terms of accessibility and safety. For example, PhET Interactive Simulations allows students to perform titrations without having to work with real chemicals. The program allows them to simulate reactions, change parameters such as solution

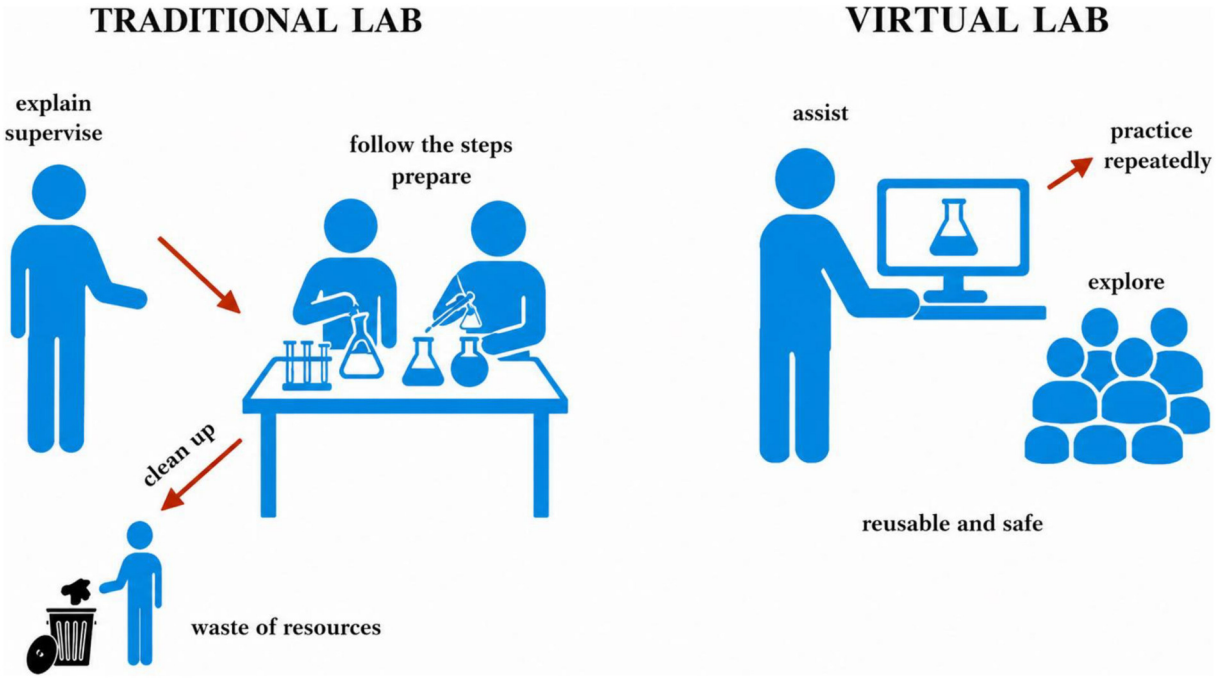


FIGURE 3
Comparison of traditional and virtual chemistry laboratory.

concentration and temperature, and observe the results (Gungor et al., 2022). Virtual labs provide the ability to repeat experiments multiple times, which helps students better understand the material and obtain accurate results without additional costs for reagents. Moreover, a virtual lab leaves no waste, which reduces the ecological footprint and simplifies the learning process. Students can easily experiment, repeat reactions under different conditions, and observe how changes affect the result, which helps them better understand chemical principles (Brown et al., 2021).

Molecular visualization tools are particularly important for supporting learning at the submicroscopic level, which is widely recognized as one of the most challenging aspects of chemistry education. Such technologies enable students to observe molecular structures, particle interactions, and dynamic reaction processes that cannot be directly perceived in traditional classroom environments. This supports the development of conceptual understanding by helping learners connect symbolic equations and macroscopic experimental observations with underlying molecular behavior.

However, virtual labs also have important limitations. First, they cannot fully replace the physical experience of working with real chemicals and laboratory equipment. Students operating in virtual environments do not directly interact with materials, which limits the development of tactile skills and practical laboratory competence (Brinson, 2015; De Jong et al., 2013). In addition, despite the high accuracy of simulations, virtual laboratories cannot always fully reproduce the complexity and variability of real chemical processes, which may reduce their practical relevance (Brinson, 2015).

Furthermore, the effectiveness of virtual laboratories is not uniform across all learning contexts. Evidence suggests that virtual labs may be less effective for novice learners who lack prior laboratory experience, as they may struggle to understand experimental procedures without hands-on practice and appropriate instructional guidance (Smetana and Bell, 2012; Zacharia et al., 2015). Similarly, students with lower levels of self-regulation or digital literacy may encounter difficulties in navigating virtual environments, which can negatively affect learning outcomes (Makransky and Petersen, 2021). Virtual laboratories may also fail to support the development of essential procedural skills, such as precise measurement, equipment handling, and experimental accuracy. In addition, in the absence of adequate teacher guidance, students may engage with simulations at a superficial level, focusing on task completion rather than developing a deep conceptual understanding of chemical processes (Smetana and Bell, 2012).

These findings indicate that virtual laboratories are most effective when used as a complementary tool rather than a full replacement for traditional laboratory instruction. Hybrid approaches that combine virtual and physical experimentation appear to provide more balanced learning outcomes by integrating conceptual understanding with practical skill development (Olympiou and Zacharia, 2012; Potkonjak et al., 2016). Such blended environments allow students to benefit from the safety and flexibility of virtual experimentation while simultaneously developing hands-on laboratory competencies in authentic settings.

Thus, virtual laboratories are becoming an important tool in modern chemistry education, providing students with access to experiments and models that cannot be implemented in a real laboratory. These technologies help overcome limitations related to the availability of laboratory equipment, safety, and costs. There are several types of virtual laboratories, each with its own

pedagogical features and opportunities for active learning. Let us consider the main types of virtual laboratories, such as chemical reaction simulations, molecular modeling, virtual reality (VR), and others, as well as their role in the educational process (Table 1).

Thus, different types of virtual labs, such as chemical reaction simulations, molecular modeling, virtual reality, and working with real data, have their own unique pedagogical features. They promote a deeper understanding of chemical processes, develop critical thinking, and allow students to work with data and solve real problems. Virtual labs also promote active learning, where students not only gain theoretical knowledge but also practice solving real chemical problems, making them an important tool in modern STEM education.

The use of virtual labs in chemistry education opens up new horizons for educators, providing innovative teaching methods and developing skills in students that are difficult to develop through traditional lab work (Richard and Rajakumari, 2023). However, for these technologies to effectively support learning, it is necessary to implement competent pedagogical approaches that will take into account the features of virtual labs and their interaction with the educational process (Table 2).

Thus, to make the most of their potential, it is important to use appropriate pedagogical approaches such as active learning, project-based learning, inclusivity, flexibility and feedback. This will not only improve the understanding of theoretical concepts but also develop the practical skills that students need to succeed in their careers in chemistry and other sciences.

The benefits of using virtual labs are:

- Accessibility and affordability. Virtual labs allow you to conduct experiments that may require expensive materials or equipment without having to pay a financial outlay. Students can work anytime and from anywhere, without depending on the availability of the lab. This is especially important in educational institutions with limited resources.
- Safety. Virtual labs provide a safe environment for conducting dangerous or complex experiments, such as working with toxic chemicals, high temperatures, or explosive reagents. Students can practice such reactions without risking their health or safety.
- Interactivity and simulation of complex processes. Virtual labs offer interactive learning where students can actively change experimental parameters, make observations, and get instant results. This helps to gain a deeper understanding of theoretical aspects of chemistry, such as molecular interactions, reaction mechanisms, or kinetics.
- Ability to repeat experiments multiple times. Virtual labs allow students to repeat experiments as many times as needed to gain deeper understanding of the material and test hypotheses. Unlike real laboratories, where there are limitations on time, materials, and equipment, virtual laboratories allow for endless repetition of experiments and varying conditions to obtain different results.
- Simulation of inaccessible experiments. Some chemistry experiments cannot be performed in a real laboratory due

TABLE 1 Categories of STEM technologies used in chemistry education.

Type of Virtual Lab	Examples	Pedagogical Features	Pedagogical Role in Chemistry Learning	Benefits	Sources
Virtual simulations and modeling	PhET Interactive Simulations	Simulations such as Molecular Motion demonstrate how molecules respond to temperature changes and allow exploration of gas, liquid, and solid behavior	Supports understanding of submicroscopic processes and links macroscopic observations with molecular-level explanations	Promotes critical thinking and conceptual understanding through parameter manipulation.	(Castro, 2025; Salame and Makki, 2021)
Virtual Reality (VR)	Labster	The platform offers virtual laboratories with elements of virtual reality, where students can interact with molecules in 3D space, simulate reactions and observe molecular changes in real time.	Enhances spatial understanding of molecular structures and reaction mechanisms	Supports deeper conceptual learning and visualization of complex systems.	(Гулай et al., 2024)

TABLE 2 Pedagogical approaches when using virtual labs.

Pedagogical approach	Description	Learning outcome	Sources
Project-based learning	Virtual labs used for solving real chemistry problems	Improves problem-solving and application of chemical knowledge in interdisciplinary contexts	(Fadli et al., 2024)
Active learning and feedback	Students conduct experiments and receive feedback	Enhances conceptual understanding and critical thinking	(Van Den Beemt et al., 2023)
Flexibility in Learning	Students can repeat experiments anytime	Supports self-paced learning and knowledge retention	(Reginald, 2023)
Inclusiveness and Accessibility	Provides access for diverse learners	Improves participation and access to laboratory experience	(Deriba et al., 2024)

to their complexity, cost, or danger. Virtual laboratories allow for the simulation of such experiments, providing students with hands-on experience in conditions that cannot be reproduced in a physical laboratory.

Thus, virtual laboratories open up significant opportunities for improving the educational process, making chemistry education more accessible and flexible. They support the development of conceptual understanding, critical thinking, and active student engagement, while enabling interaction with complex chemical processes that may be difficult to observe in traditional settings. At the same time, the analysis indicates that their effectiveness depends on pedagogical design, student readiness, and the learning context. Virtual laboratories are less effective when used in isolation or without sufficient instructional support, particularly for developing practical laboratory skills. Therefore, a balanced approach that combines virtual and physical laboratories appears to provide the most effective learning outcomes, integrating conceptual understanding with hands-on experience.

4.3 Mathematical modeling and calculation of chemical processes

The reviewed literature also identifies mathematical modeling as an important category of STEM-related technology in chemistry education (RQ1). Its pedagogical role is primarily associated with supporting conceptual understanding of dynamic chemical systems and promoting model-based reasoning and inquiry-oriented learning processes (RQ2).

One of the most important aspects of using STEM technologies in chemical education is the use of mathematical modeling and calculations of chemical processes (Wisudawati, 2018). These methods allow not only to analyze complex chemical reactions, but also to predict their results when changing various parameters (Prishchenko et al., 2022; Schuchardt and Schunn, 2016). From an educational perspective, mathematical modeling should be viewed not only as a tool for scientific analysis but also as a powerful means of supporting conceptual understanding in chemistry learning. Research indicates that engaging students in modeling activities enhances their ability to understand dynamic chemical processes, such as reaction kinetics and equilibrium, by linking mathematical representations with underlying chemical concepts. In teaching practice, modeling is commonly implemented through guided problem-solving, simulation-based tasks, and inquiry-based learning environments, where students construct, test, and refine models rather than simply apply predefined formulas.

In chemistry education, mathematical modeling should not be viewed solely as a computational or technical activity. Rather, it functions as a pedagogical tool that supports students' understanding of dynamic chemical relationships, including reaction kinetics, equilibrium, and thermodynamic dependencies. When combined with visualization and simulation technologies, mathematical modeling helps students interpret abstract symbolic representations in relation to observable chemical phenomena.

At the same time, mathematical modeling of chemical processes begins with choosing the right model that will most accurately describe the behavior of the system. These models can be based on areas such as chemical kinetics, thermodynamics, molecular dynamics, and others. The main task is to create equations that can predict the behavior of chemicals when external conditions such as temperature, pressure, or concentration of reactants change. Models can be simple, for example, to describe elementary reactions, or complex, when it is necessary to take into account many factors, such as multicomponent reactions or the interaction of molecules at the atomic level (Serovajsky, 2021). It is important to note that mathematical modeling in chemistry allows you to obtain results that cannot always be measured experimentally, and helps to optimize various processes.

One of the most common applications of mathematical modeling in chemistry is reaction kinetics (Rice et al., 2023). Mathematical modeling allows us to predict how the rate of a reaction will change when factors such as temperature, pressure, or reactant concentration are changed. This is especially important for industrial chemical processes, where reaction rates must be precisely controlled to achieve maximum efficiency (Rivera et al., 2021). When producing chemicals in large plants, it is important to know how changes in temperature or reactant concentration will affect the rate of a reaction. Modeling the rate of chemical reactions is important not only for fundamental research, but also for industrial applications where reactions need to be controlled to ensure they proceed with maximum efficiency. For example, in the production of ammonia using the Haber-Bosch method, chemists can model how changes in temperature or reactant concentrations will affect the reaction rate (Moghadam et al., 2024). This helps to select the optimal conditions under which the reaction will proceed most efficiently, with minimal expenditure of energy and reactants. In addition, mathematical modeling allows for the optimization of reaction conditions in laboratories and production facilities, improving productivity and reducing costs.

Another important aspect of chemical processes that can be modeled is chemical equilibrium. In chemistry, equilibrium is a state in which the rates of the forward and reverse reactions become equal and the concentrations of the reactants remain constant. Mathematical modeling of chemical equilibrium allows us to predict which species will predominate in a reaction under various conditions, such as temperature, pressure, and reactant concentration (Rasmuson, 2014).

Equilibrium modeling is especially important in the design of processes such as chemical synthesis or waste recycling. For example, in the process of synthesizing ammonia, the equilibrium of the reaction between nitrogen and hydrogen must be controlled (Mirvakili et al., 2021). Mathematical modeling helps chemists calculate at what temperatures and pressures the reaction will be in optimal equilibrium to maximize the yield of ammonia. Using mathematical models to predict chemical equilibrium can significantly improve the efficiency of processes, minimizing costs and resources, and reducing waste.

However, research also indicates that students often experience significant difficulties when working with mathematical models in chemistry. These challenges are related

to the abstract nature of symbolic representations, limited mathematical background, and difficulties in connecting equations with chemical meaning. As a result, students may focus on procedural calculations without fully understanding the underlying chemical processes. This highlights the need for instructional approaches that integrate visualization tools, scaffolding, and step-by-step model construction to support meaningful learning.

Catalysis is a process in which the rate of a chemical reaction is increased by a catalyst, but the catalyst itself is not consumed in the reaction. Mathematical modeling of catalysis helps us understand how catalysts affect the rate of a reaction and how their effectiveness depends on various factors, such as temperature, pressure, and reactant composition (Ivanchina et al., 2021). Models of catalysis based on mathematical principles can take into account factors such as the adsorption of molecules on the catalyst surface and the activation energy of the reaction. This modeling helps select the best catalysts for specific reactions and optimize the conditions for their effective use. For example, in the petrochemical industry, catalysts are used to accelerate the hydrogenation reaction, converting unsaturated hydrocarbons into saturated ones (Mahmoudi et al., 2024). Mathematical modeling allows us to predict how different catalysts will interact with reactants, which helps in developing more efficient and environmentally friendly methods of hydrocarbon processing.

For complex calculations of chemical processes, chemists use specialized software packages such as Matlab, Mathematica, COMSOL, and Gaussian (Idoko et al., 2024; Teixeira-Dias, 2017). These programs allow you to model chemical reactions taking into account many factors and help analyze the results obtained during experiments. Programs of this kind are used to predict the behavior of molecules, simulate interactions at the atomic level, and optimize conditions for chemical reactions. For example, COMSOL Multiphysics is software that helps simulate chemical reactions in real time, as well as optimize chemical processes such as pyrolysis, catalysis, and material synthesis (Wijayanti et al., 2021).

Using such software allows chemists to develop new technologies faster and more accurately, as well as analyze complex chemical processes that cannot be carried out experimentally due to their complexity or cost.

The future of mathematical modeling in chemistry is associated with the development of computing technologies, artificial intelligence, and machine learning (He et al., 2021; Karthikeyan and Priyakumar, 2022; Staszak, 2023). These new technologies will allow the creation of more complex and accurate models of chemical processes that will take into account all microscopic interactions at the atomic level.

The integration of artificial intelligence further expands the potential of mathematical modeling by enabling the design of molecules with predefined properties and accelerating the development of new materials. While these advances are primarily associated with research and industrial applications, they also have important implications for education, as they introduce new opportunities for data-driven and simulation-based learning.

Thus, mathematical modeling plays a dual role in chemistry education and practice. While it remains essential for analyzing and optimizing chemical processes in research and industry, its pedagogical value lies in supporting students' ability to understand complex systems and relationships. The literature suggests that modeling is most effective when integrated into learning as an active process involving interpretation, prediction, and reflection, rather than as a purely computational task. At the same time, its successful implementation requires appropriate instructional design, including scaffolding, visualization, and alignment with students' prior knowledge.

Overall, the evidence indicates that mathematical modeling should be viewed not only as a technical skill but as a key component of conceptual learning in STEM-based chemistry education, particularly when combined with interactive technologies and guided instructional approaches.

Taken together, the reviewed studies indicate that the main categories of STEM-related technologies in chemistry education include virtual laboratories, simulation environments, molecular visualization systems, and mathematical modeling tools, thereby addressing RQ1. Across the reviewed literature, these technologies were found to support conceptual understanding and inquiry-oriented learning by facilitating visualization of abstract chemical processes, interactive experimentation, and model-based reasoning, thereby addressing RQ2. These tools enhance conceptual understanding by enabling visualization of abstract chemical processes and by allowing students to explore relationships between variables in dynamic environments. At the same time, the analysis reveals notable differences and limitations. The effectiveness of STEM technologies varies depending on students' prior knowledge, level of self-regulation, and digital competence. In addition, while virtual and simulation-based tools are particularly effective in supporting conceptual learning, they are less successful in developing practical laboratory skills when used in isolation. Furthermore, variations in teacher preparedness and institutional resources contribute to differences in implementation outcomes across studies. These findings suggest that the successful integration of STEM technologies requires not only access to digital tools but also carefully designed pedagogical strategies that align with the cognitive and disciplinary demands of chemistry education.

Across the reviewed studies, STEM-related technologies were found to be most effective when they supported chemistry-specific learning processes, particularly students' ability to connect macroscopic observations, symbolic representations, and submicroscopic explanations. The literature consistently indicates that visualization, simulation, and modeling tools play a central role in reducing conceptual difficulties associated with abstract chemical phenomena and in supporting meaningful conceptual understanding in chemistry education.

5 Problems and challenges of implementing STEM technologies

This section addresses RQ3 by analyzing the major challenges, limitations, and future directions associated with the implementation of STEM technologies in chemistry education.

The integration of STEM technologies into chemistry education is a powerful tool for modernizing the educational process. The introduction of science, technology, engineering,

and mathematics into educational programs allows for the creation of more practice-oriented and interdisciplinary education that is aimed at solving real problems of the modern world. Integrating STEM technologies into chemistry education has a number of significant advantages. Firstly, it promotes the development of interdisciplinary skills (Eugenijus, 2023), as students study chemistry in the context of other disciplines, such as mathematics, engineering and technology. This helps them see the relationship between different fields of science and develops the ability to solve complex problems. Secondly, the STEM approach increases the practical focus of learning (Juškevičienė et al., 2021). Students participate in real-life projects, which allows them not only to assimilate theoretical knowledge, but also to apply it in practice to solve current problems, such as the creation of environmentally friendly technologies or innovative materials. In addition, the introduction of STEM technologies promotes the development of teamwork skills, as many projects require students to interact with various scientific and technical specialists (Fang et al., 2021; Owens and Hite, 2022). This improves their ability to work in interdisciplinary teams, which is an important skill in the modern labor market.

The STEM approach also prepares students to address global challenges such as climate change and the energy crisis by offering them the opportunity to develop solutions that can impact the environment, energy, and medicine (Al Hamad et al., 2024). In addition, this approach promotes creativity and innovative thinking, helping students generate new ideas and find unconventional solutions to existing problems. Although the integration of STEM technologies into chemistry education has many advantages, there are also certain problems and challenges that need to be taken into account. These difficulties can arise both at the level of educational institutions and at the level of teachers and students. The implementation of STEM in the educational process requires significant efforts to overcome these obstacles, but it can also lead to a significant improvement in the quality of education and the preparation of students for modern challenges.

The implementation of STEM technologies in chemistry education is associated with a number of disadvantages (Acevedo et al., 2024). Firstly, there is a shortage of qualified teachers who can effectively teach interdisciplinary courses that require knowledge not only in chemistry, but also in mathematics, engineering and technology (Hamad et al., 2022). This creates difficulties in training highly qualified specialists in these fields. Secondly, the successful implementation of the STEM approach requires significant funding for updating the curriculum, purchasing modern equipment and software (Adebusuyi et al., 2022). In some educational institutions, especially with a limited budget, such costs can be unaffordable (Uludüz and Çalik, 2022).

In addition, students may find it difficult to adapt to the new approach, especially if they are accustomed to traditional teaching methods. The transition from theoretical learning to project-based and interdisciplinary learning may cause them certain difficulties (Sahin, 2019; Tantayanon et al., 2024). Also, in some countries and educational systems there may be resistance to the introduction of new teaching methods, which slows down the process of integrating STEM technologies into curricula. In order to clearly and systematically assess the advantages and challenges of implementing STEM technologies in chemical education, a SWOT analysis was conducted. This analysis helps to identify the strengths and weaknesses of this approach, as well as the opportunities and threats that educational institutions may face when integrating it. Understanding these factors is important in order to effectively implement STEM technologies and minimize possible risks (Table 3).

The SWOT analysis indicates that challenges in implementing STEM technologies in chemistry education are not uniform and vary depending on context. Structural factors, such as teacher expertise and curriculum flexibility, play a central role in determining the effectiveness of implementation. At the same time, context-dependent factors, including access to digital infrastructure and institutional resources, significantly influence the scalability of STEM approaches. In addition, discipline-specific challenges related to the abstract nature of chemistry, particularly the need to connect macroscopic, submicroscopic, and symbolic representations, create additional cognitive demands for learners. These findings suggest that successful integration of STEM technologies requires not only technological investment but also targeted pedagogical strategies adapted to the specific characteristics of chemistry education.

Overall, the analysis shows that the effectiveness of STEM technologies in chemistry education depends on the interaction of structural, contextual, and discipline-specific factors. In particular, the need to integrate macroscopic, submicroscopic, and symbolic representations creates additional cognitive challenges for learners. Therefore, successful implementation requires not only technological resources, but also carefully designed pedagogical strategies adapted to the specific nature of chemistry learning.

However, the implementation of STEM in the educational process also faces a number of challenges. These include a lack of qualified teachers who can effectively teach interdisciplinary courses, high financial costs for updating teaching materials and equipment, as well as difficulties in adapting students to a new learning style. In addition, some educational systems may encounter resistance to change, which will hinder the implementation of innovative methods. Nevertheless, the opportunities that the STEM approach opens up for education, such as cooperation with industry, the development of new educational technologies, and improving students' career prospects, create significant potential for its successful integration into educational programs. Successful overcoming of the identified threats and weaknesses, such as problems with access to resources or the need for constant updating of programs, will contribute to the full implementation of STEM technologies and improve the quality of education in chemistry.

Overall, the findings related to RQ3 demonstrate that the implementation of STEM technologies in chemistry education is influenced by multiple structural, pedagogical, and contextual factors, including teacher preparedness, technological infrastructure, curriculum flexibility, and students' readiness for technology-enhanced learning.

6 Recommendations for improving the integration of STEM technologies

The following recommendations are derived from the analysis of the reviewed literature presented in Sections 4 and 5. In

TABLE 3 SWOT analysis of the implementation of STEM technologies in chemical education.

Factor	Strengths	Weaknesses
Strengths	1. Support for multilevel chemical understanding: STEM tools (simulations, molecular visualization) help link macroscopic, submicroscopic, and symbolic representations. 2. Visualization of abstract chemical processes: Enables understanding of molecular interactions and reaction mechanisms that cannot be directly observed. 3. Safe exploration of hazardous chemical experiments: Allows simulation of dangerous reactions without risk. 4. Integration of real-world chemical applications: Connects chemistry with environmental, industrial, and materials science problems.	1. High demands on teacher expertise: Requires interdisciplinary knowledge and digital competence. 2. High cost of chemistry-specific technologies: Virtual labs, sensors, and modeling software require significant investment, especially in resource-limited settings. 3. Limited development of practical laboratory skills: Lack of hands-on experience with real chemicals and equipment. 4. Student difficulties in interpreting models: Challenges in linking equations and simulations to chemical meaning.
Weaknesses	1. Difficulties integrating STEM into existing chemistry curricula due to limited time and rigid course structures.	1. Need for continuous modernization of laboratory infrastructure and digital tools.
Opportunities	1. Expansion of virtual laboratories and simulation-based learning in chemistry education. 2. Development of data-driven and modeling-based learning (e.g., kinetics and equilibrium modeling). 3. Strengthening collaboration with industry in areas such as materials science, environmental chemistry, and chemical engineering.	1. Unequal access to chemistry-specific technologies across regions and institutions. 2. Dependence on external funding for sustainable implementation. 3. Resistance to pedagogical change in traditional chemistry education systems.
Threats	1. Rapid evolution of technologies may outpace teachers' ability to integrate them effectively. 2. Increasing gap between technologically advanced and resource-limited institutions.	1. Limited curricular flexibility to incorporate modeling and interdisciplinary STEM activities. 2. Risk of over-reliance on simulations, reducing development of experimental competence.

particular, they reflect key findings related to the importance of pedagogical design, student difficulties in understanding abstract chemical concepts, and the role of STEM technologies in supporting conceptual learning. These recommendations aim to address the structural, contextual, and discipline-specific challenges identified in the analysis.

The integration of STEM technologies into chemical education opens up new opportunities to improve the quality of education, but for effective implementation there are certain problems that need to be addressed. Based on the reviewed studies, one of the key steps to effectively integrating STEM technologies into chemistry education is to develop interdisciplinary courses that connect chemistry with mathematics, engineering, and technology (Dyachenko et al., 2024). For example, courses could be created that incorporate chemical reaction theory, mathematical modeling techniques, and engineering fundamentals, helping students see how all of these disciplines interact to solve real-world problems (Nugraha et al., 2024). For example, the article describes an approach to integrating humanities into chemistry instruction through a course modification. Changes were made to the curriculum structure, including historical contexts and references to mathematics. Students were given assignments that encouraged them to connect scientific concepts with humanities subjects. Perception data were collected using a Likert-scale questionnaire before and after the course implementation. The results showed that most students reported an improved understanding of the connections between science and other disciplines, as well as between science and their everyday lives. To facilitate students' transition to interdisciplinary learning environments that incorporate mathematical modeling, instructors should implement scaffolded learning strategies. These may include guided model construction, worked examples, visual representations of chemical processes, and a gradual progression from simple to more complex modeling tasks. Such support can help reduce cognitive overload, strengthen connections between mathematical representations and chemical concepts, and improve students' confidence when working with STEM-based technologies. The integration increased interest in learning, deepened understanding of chemistry, and emphasized its relevance in a cultural context. Overall, the integrated approach contributed to the development of interdisciplinary perspectives in students and improved connections between different academic fields (Doğru, 2023).

The analysis of the literature indicates that for successful implementation of the STEM approach in chemical education, it is necessary for teachers to have interdisciplinary knowledge and be able to teach not only chemistry but also other areas of science and technology. It is important to organize professional development programs for teachers so that they can master new educational methods and technologies, such as the use of virtual laboratories and online courses. An example is a professional development course for chemistry teachers on the use of virtual chemistry laboratories and interactive simulators, such as ChemCollective, which can be integrated into the educational process (Kriscautzky Laxague, 2024).

As highlighted in several studies, ensuring the relevance of educational programs and increase their practical value, it is important to develop cooperation with industry and research institutions. For example, universities can enter into

partnerships with companies working in the field of biotechnology, chemical industry and new materials so that students can participate in real projects and internships. This will help students not only to study theoretical aspects but also to apply them in practice. An example could be a joint project with chemical plants to develop new materials with improved characteristics (Kaya-Capocci et al., 2025). The paper describes a successful collaboration between a high school and a local STEM company that resulted in significant educational outcomes. The partnership created a real-life learning environment where students not only learned STEM subjects in depth, but also had the opportunity to apply their knowledge in practice through industry projects. This collaboration gave students the opportunity to work on real-life problems, which increased their interest and engagement in further education and careers in STEM. One of the key success factors was overcoming cultural and communication barriers between the educational institution and industry. The partners were able to effectively interact, which contributed to the creation of trusting relationships and mutual support. An important outcome of this collaboration was the increased professional development of both students and teachers. The principles used to develop the partnership included a willingness to consider the needs of both parties, building trust, and using leadership support, which ensured the sustainability and effectiveness of the partnership. Thus, the results of the partnership demonstrated the importance of having a clear strategy, flexibility in approach, and the involvement of leaders from both sides to create an environment that promotes STEM education and supports the professional growth of students and teachers (Perry, 2023).

The findings also suggest that personalized learning approaches using online courses and virtual labs allows students to learn at their own pace. It is important to integrate platforms like Coursera, edX, and Khan Academy into the learning process so that students can explore additional STEM topics in depth. This will also help teachers work with each student on an individual basis, and students can explore additional topics that interest them or that they are struggling with (Chasokela, 2024; Ngendabanga et al., 2025).

Overall, the reviewed evidence indicates that the effective integration of STEM technologies in chemistry education requires a strong alignment between technological tools and pedagogical strategies. In particular, emphasis should be placed on supporting students' understanding of abstract chemical concepts, providing appropriate scaffolding, and ensuring that digital tools complement rather than replace experimental practice. These findings highlight that successful implementation depends not only on access to technology but also on the development of discipline-specific teaching approaches.

6.1 Limitations of the review

This review has several limitations that should be considered when interpreting the findings. First, it is based on a narrative review approach, which does not follow a strict systematic protocol and may therefore be subject to selection bias. Although efforts were made to include relevant and recent studies, the selection process may not be fully exhaustive or reproducible.

Second, the analysis primarily focuses on peer-reviewed publications, which may introduce publication bias by underrepresenting studies with negative or inconclusive results. Third, the reviewed studies vary considerably in terms of educational context, including differences between secondary and higher education, as well as between developed and resource-limited settings. This heterogeneity limits the generalizability of the findings.

Finally, although the review emphasizes literature published between 2013 and 2025, a limited number of earlier studies were included due to their conceptual relevance to STEM education. These limitations suggest that the findings should be interpreted with caution and highlight the need for further systematic and empirical research in this area.

7 Conclusions

This review provides a structured analysis of the integration of STEM technologies in chemistry education by linking technological tools with their pedagogical functions and learning outcomes. Unlike previous studies that focus on individual technologies, this review offers a cross-study synthesis highlighting how virtual laboratories, simulations, and mathematical modeling support conceptual understanding of chemical processes.

The findings indicate that the effectiveness of STEM technologies depends not only on their availability but also on pedagogical design, particularly the use of scaffolding, inquiry-based learning, and alignment with the multilevel nature of chemical knowledge (macroscopic, submicroscopic, and symbolic representations). At the same time, the analysis reveals key limitations, including difficulties in developing practical laboratory skills, variability in student readiness, and the influence of contextual factors such as teacher expertise and resource availability.

Importantly, this review demonstrates that no single technology is sufficient to address the complexity of chemistry learning. Instead, integrated approaches combining digital tools with traditional laboratory experiences provide the most effective outcomes.

Overall, the findings of this review address the three research questions proposed in the study. Regarding RQ1, the analysis identified virtual laboratories, simulation environments, molecular visualization systems, and mathematical modeling tools as the main categories of STEM-related technologies applied in chemistry education. In relation to RQ2, the reviewed studies demonstrated that these technologies support conceptual understanding and inquiry-oriented learning by helping students connect macroscopic observations, symbolic representations, and submicroscopic explanations of chemical phenomena. Regarding RQ3, the review revealed that the effective implementation of STEM technologies depends on multiple pedagogical and contextual factors, including teacher preparedness, technological infrastructure, curriculum flexibility, and students' readiness for technology-enhanced learning.

Future research should focus on developing discipline-specific pedagogical frameworks for integrating STEM technologies in chemistry education, with particular attention to supporting students' understanding of abstract chemical concepts. In

addition, further empirical studies are needed to examine how different combinations of technologies influence learning outcomes across educational contexts, including secondary and higher education settings.

Overall, the results suggest that the successful integration of STEM technologies in chemistry education requires not only technological innovation but also carefully designed instructional strategies that reflect the cognitive and disciplinary complexity of chemistry.

Author contributions

BM: Conceptualization, Methodology, Project administration, Visualization, Writing – original draft. AD: Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Visualization, Writing – original draft, Writing – review & editing. IA: Data curation, Investigation, Resources, Validation, Writing – original draft. ZD: Investigation, Software, Visualization, Writing – review & editing. LZ: Formal analysis, Investigation, Methodology, Visualization, Writing – original draft. RZ: Investigation, Software, Writing – original draft. BI: Data curation, Investigation, Resources, Writing – review & editing. BS: Conceptualization, Visualization, Writing – original draft. AU: Formal analysis, Methodology, Writing – original draft. AK: Data curation, Formal analysis, Investigation, Writing – original draft. KA: Conceptualization, Data curation, Validation, Writing – review & editing. MA: Formal analysis, Investigation, Resources, Writing – original draft.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Acknowledgments

The author expresses gratitude to colleagues who provided valuable comments on earlier versions of the manuscript.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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References

Abungu, H. E., Okere, M. I. O., and Wachanga, S. W. (2014). The effect of science process skills teaching approach on secondary school Students' achievement in chemistry in Nyando District, Kenya. *JESR.* 4 (6), 359–372. doi: 10.5901/jesr.2014.v4n6p359

Acevedo, P., Magana, A. J., Benes, B., and Mousas, C. (2024). A systematic review of immersive virtual reality in STEM education: advantages and disadvantages on learning and user experience. *IEEE. Access.* 12, 189359–189386. doi: 10.1109/ACCESS.2024.3489233

Adebusuyi, O. F., Bamidele, E. F., and Adebusuyi, A. S. (2022). The role of knowledge and epistemological beliefs in chemistry teachers STEM professional development and instructional practices: examination of STEM-integrated classrooms. *Eur. J. Sci. Math. Ed.* 10, 243–255. doi: 10.30935/scimath/11799

Akinola, Y. M., Agbonifo, O. C., and Sarumi, O. A. (2020). Virtual reality as a tool for learning: the past, present and the prospect. *J. Appl. Learn. Teach.* 3, 51–58. doi: 10.37074/jalt.2020.3.2.10

Akrami, Z. (2022). The effectiveness of education with the STEM approach in the development of entrepreneurial thinking in chemistry students. *Chem. Educ. Res. Pract* 23, 475–485. doi: 10.1039/D2RP00011C

Al Hamad, N. M., Adewusi, O. E., Unachukwu, C. C., Osawaru, B., and Chisom, O. N. (2024). A review on the innovative approaches to STEM education. *Int. J. Sci. Res. Arch* 11, 244–252. doi: 10.30574/ijsra.2024.11.1.0026

Alhashem, F., and Alfailakawi, A. (2023). Technology-enhanced learning through virtual laboratories in chemistry education. *Cont. Ed. Technology.* 15, ep474. doi: 10.30935/cedtech/13739

Ambrož, M., Pernaa, J., Haatainen, O., and Aksela, M. (2023). Promoting STEM education of future chemistry teachers with an engineering approach involving single-board computers. *Appl. Sci.* 13, 3278. doi: 10.3390/app13053278

Braun, V., and Clarke, V. (2006). Using thematic analysis in psychology. *Qual. Res. Psychol.* 3, 77–101. doi: 10.1191/1478088706qp063oa

Brinson, J. R. (2015). Learning outcome achievement in non-traditional (virtual and remote) versus traditional (hands-on) laboratories: a review of the empirical research. *Comput. Educ.* 87, 218–237. doi: 10.1016/j.compedu.2015.07.003

Brown, C. E., Alrmuny, D., Williams, M. K., Whaley, B., and Hyslop, R. M. (2021). Visualizing molecular structures and shapes: a comparison of virtual reality, computer simulation, and traditional modeling. *Chem. Teach. Int.* 3, 69–80. doi: 10.1515/cti-2019-0009

Bybee, R. W. (2013). *The Case for STEM Education: Challenges and Opportunities.* Arlington, VA: National Science Teachers Association.

Castro, R. (2025). The effects of chemistry virtual laboratories in academic achievement of secondary level learners: a meta-analysis. *In. Sci. Ed. J.* 6, 24–37. doi: 10.37251/isej.v6i1.1379

Chan, P., Van Gerven, T., Dubois, J.-L., and Bernaerts, K. (2021). Virtual chemical laboratories: a systematic literature review of research, technologies and instructional design. *Comput. Educ. Open* 2, 100053. doi: 10.1016/j.caeo.2021.100053

Chasokela, D. (2024). "Framework for implementing STEM learning technologies: addressing students' challenges of developing 21st-century skills," in *Advances in Higher Education and Professional Development*, eds. M. Kayyali, and B. Christiansen (Hershey, PA: IGI Global), 169–208. doi: 10.4018/979-8-3693-3443-0.ch007

Clucas, P., and Sjöström, J. (2024). Traces of bildung in upper secondary science education: a critical investigation of chemistry teachers' orientation towards promoting bildung in chemistry education. *Sci. Educ.* 34, 2225–2250. doi: 10.1007/s11191-024-00510-6

De Jong, T., Linn, M. C., and Zacharia, Z. C. (2013). Physical and virtual laboratories in science and engineering education. *Science.* 340, 305–308. doi: 10.1126/science.1230579

Deriba, F. G., Saqr, M., and Tukiainen, M. (2024). Assessment of accessibility in virtual laboratories: a systematic review. *Front. Educ.* 9, 1351711. doi: 10.3389/feduc.2024.1351711

Doğru, S. (2023). Integration of STEM education to humanities: examining interdisciplinary links in basic chemistry course according to student views. *Res. Educ. Psychol.* 7, 128–139. doi: 10.54535/rep.1340404

Dyachenko, A., Mukanova, R., and Erkibayeva, M. (2024). Integration of interdisciplinary connections between chemistry, physics, and biology in the education of secondary school students. *Int. J. Educ. Reform*, doi: 10.1177/10567879241290180

Eugenijus, L. (2023). Integrating blended learning and STEM education: innovative approaches to promote interdisciplinary learning. *RAE.* 2, 20–36. doi: 10.56397/RAE.2023.09.03

Fabeku, O. E., and Enyeasi, S. C. (2024). Computer simulation and video media instructional packages in improving learning outcomes of chemistry students in Ife central local government area. *Discip. Interdscip. Sci. Educ. Res.* 6, 19. doi: 10.1186/s43031-024-00109-5

Fadli, R., Surjono, H. D., Sari, R. C., Eliza, F., Hakiki, M., Hidayah, Y., et al. (2024). Effectiveness of mobile virtual laboratory based on project-based learning to build constructivism thinking. *Int. J. Interact. Mob. Technol* 18, 40–55. doi: 10.3991/ijim.v18i06.47643

Fang, M., Jandigulov, A., Snezhko, Z., Volkov, L., and Dudnik, O. (2021). New technologies in educational solutions in the field of STEM: the use of online communication services to manage teamwork in project-based learning activities. *Int. J. Emerg. Technol. Learn.* 16, 4–22. doi: 10.3991/ijet.v16i24.25227

Faulconer, E. K., Griffith, J. C., Wood, B. L., Acharyya, S., and Roberts, D. L. (2018). A comparison of online and traditional chemistry lecture and lab. *Chem. Educ. Res. Pract.* 19, 392–397. doi: 10.1039/C7RP00173H

Ferrari, R. (2015). Writing narrative style literature reviews. *Med. Writing.* 24, 230–235. doi: 10.1179/2047480615Z.000000000329

Fitriyana, N., Wiyarsi, A., Pratomo, H., and Marfuatun, M. (2024). The importance of integrated STEM learning in chemistry lesson: perspectives from high school and vocational school chemistry teachers. *J. Technol. Sci. Educ.* 14, 418. doi: 10.3926/jotse.2356

Gilbert, J. K., and Treagust, D. (2009). *Multiple Representations in Chemical Education*. Dordrecht: Springer Netherlands. doi: 10.1007/978-1-4020-8872-8

Gungor, A., Kool, D., Lee, M., Avraamidou, L., Eisink, N., Albada, B., et al. (2022). The use of virtual reality in a chemistry lab and its impact on students' self-efficacy, interest, self-concept and laboratory anxiety. *EURASIA J. Math Sci. Tech. Ed.* 18, em2090. doi: 10.29333/ejmste/11814

Habiddin, H., and Nagol, I. L. (2023). Chemistry students' mathematics ability and their understanding of buffer solution. *J. Penelit. Pendidik. IPA.* 9, 8140–8145. doi: 10.29303/jppipa.v9i10.3682

Hamad, S., Tairab, H., Wardat, Y., Rabbani, L., AlArabi, K., Yousif, M., et al. (2022). Understanding science teachers' implementations of integrated STEM: teacher perceptions and practice. *Sustainability.* 14, 3594. doi: 10.3390/su14063594

He, L., Bai, L., Dionysiou, D. D., Wei, Z., Spinney, R., Chu, C., et al. (2021). Applications of computational chemistry, artificial intelligence, and machine learning in aquatic chemistry research. *Chem. Eng. J.* 426, 131810. doi: 10.1016/j.cej.2021.131810

Hu-Au, E., and Okita, S. (2021). Exploring differences in student learning and behavior between real-life and virtual reality chemistry laboratories. *J. Sci. Educ. Technol.* 30, 862–876. doi: 10.1007/s10956-021-09925-0

Ibrahim, M., and Mohd Badli, N. A. (2025). STEM In chemistry: cultivating problem-solving skills via blended problem-based learning socratics module. *Lumat.* 12, 4. doi: 10.31129/LUMAT.12.4.2445

Idoko, P. I., Ezeamii, G. C., Idogho, C., Peter, E., Obot, U. S., and Iguoba, V. A. (2024). Mathematical modeling and simulations using software like MATLAB, COMSOL and Python. *Magna Sci. Adv. Res. Rev.* 12, 062–095. doi: 10.30574/msarr.2024.12.2.0181

Ivanchina, E. D., Ivashkina, E. N., Dolganova, I. O., and Belinskaya, N. S. (2021). Mathematical modeling of multicomponent catalytic processes of petroleum refining and petrochemistry. *Rev. Chem. Eng.* 37, 163–191. doi: 10.1515/revce-2018-0038

Jensen, L., and Konradsen, F. (2018). A review of the use of virtual reality head-mounted displays in education and training. *Educ. Inf. Technol.* 23, 1515–1529. doi: 10.1007/s10639-017-9676-0

Jiang, G., Xia, X., Li, Y., Liang, H.-N., and Hui, P. (2024). "ChemistryVR: enhancing educational experiences through virtual chemistry lab simulations," in *SIGGRAPH Asia 2024 Educator's Forum*, (Tokyo Japan: ACM), 1–5. doi: 10.1145/3680533.3697068

Johnstone, A. H. (1991). Why is science difficult to learn? Things are seldom what they seem. *J. Comput. Assist. Learn.* 7, 75–83. doi: 10.1111/j.1365-2729.1991.tb00230.x

Juškevičienė, A., Dagienė, V., and Dolgopolovas, V. (2021). Integrated activities in STEM environment: methodology and implementation practice. *Comput. Appl. Eng.* 29, 209–228. doi: 10.1002/cae.22324

Karthikeyan, A., and Priyakumar, U. D. (2022). Artificial intelligence: machine learning for chemical sciences. *J. Chem. Sci.* 134, 2. doi: 10.1007/s12039-021-01995-2

Kaya-Capocci, S., Pabuccu-Akis, A., and Orhan-Ozteber, N. (2025). Entrepreneurial STEM education: enhancing students' resourcefulness and problem-solving skills. *Res. Sci. Educ.* 55, 103–134. doi: 10.1007/s11165-024-10189-y

Kelley, T. R., and Knowles, J. G. (2016). A conceptual framework for integrated STEM education. *IJ STEM Ed.* 3, 11. doi: 10.1186/s40594-016-0046-z

Kolil, V. K., and Achuthan, K. (2024). Virtual labs in chemistry education: a novel approach for increasing student's laboratory educational consciousness and skills. *Educ. Inf. Technol.* 29, 25307–25331. doi: 10.1007/s10639-024-12858-x

Kriscautzky Laxague, M. (2024). *Didactics as the Foundation for Technological Development: The Case of the Virtual Chemistry Laboratory at Unam.,* (Valencia, Spain: IATED Academy). 6388–6394. doi: 10.21125/inted.2024.1671

Li, Y., Wang, K., Xiao, Y., Froyd, J. E., and Nite, S. B. (2020). Research and trends in STEM education: a systematic analysis of publicly funded projects. *IJ STEM Ed.* 7, 17. doi: 10.1186/s40594-020-00213-8

Mahmoudi, H., Bazrafshan, M., and Ahmadipourroudposht, M. (2024). A hybrid model of mathematical modeling and simulation for improving the petrochemical products transmission in a green supply chain. *J. Syst. Sci. Syst. Eng*, doi: 10.1007/s11518-024-5630-z

Makransky, G., and Petersen, G. B. (2021). The cognitive affective model of immersive learning (CAMIL): a theoretical research-based model of learning in immersive virtual reality. *Educ. Psychol. Rev.* 33, 937–958. doi: 10.1007/s10648-020-09586-2

Manyilizu, M. C. (2023). Effectiveness of virtual laboratory vs. paper-based experiences to the hands-on chemistry practical in Tanzanian secondary schools. *Educ. Inf. Technol.* 28, 4831–4848. doi: 10.1007/s10639-022-11327-7

Martín-Páez, T., Aguilera, D., Perales-Palacios, F. J., and Vílchez-González, J. M. (2019). What are we talking about when we talk about STEM education? A review of literature. *Sci. Educ.* 103, 799–822. doi: 10.1002/sce.21522

Mirvakili, A., Eksiri, Z., Biniaz, P., and Mohaghegh, N. (2021). Two-dimensional mathematical modeling of an industrial ammonia synthesis reactor with CFD analysis. *J. Taiwan Inst. Chem. Eng.* 121, 1–19. doi: 10.1016/j.jtice.2021.03.032

Moghadam, M. R., Bazmandegan-Shamili, A., and Bagheri, H. (2024). "The current methods of ammonia synthesis by Haber-Bosch process," in *Progresses in Ammonia: Science, Technology and Membranes*, (Amsterdam: Elsevier), 1–32. doi: 10.1016/B978-0-323-88516-4.00002-0

Nehru, N., Purwaningsih, S., Riantoni, C., Ropawandi, D., and Novallyan, D. (2024). Mapping students' thinking systems in critical thinking based on STEM project-based learning experiences. *Jur. Ilmh. Ilm. Ter. Un. Ja.* 8, 136–144. doi: 10.22437/jiituj.v8i1.32027

Ngendabanga, C., Nkurunziza, J. B., and Mugabo, L. R. (2025). Innovative approaches in chemistry teaching: a systematic review on the use of improvised chemicals for student engagement and performance. *Chem. Educ. Res. Pract.* 26, 566–577.doi: 10.1039/D4RP00302K

Nugraha, M. G., Kidman, G., and Tan, H. (2024). Interdisciplinary STEM education foundational concepts: implementation for knowledge creation. *EURASIA J. Math Sci. Tech. Ed.* 20, em2523. doi: 10.29333/ejmste/15471

Olympiou, G., and Zacharia, Z. C. (2012). Blending physical and virtual manipulatives: an effort to improve students' conceptual understanding through science laboratory experimentation. *Sci. Educ.* 96, 21–47. doi: 10.1002/sce.20463

Owens, A. D., and Hite, R. L. (2022). Enhancing student communication competencies in STEM using virtual global collaboration project based learning. *Res. Sci. Technol. Educ.* 40, 76–102. doi: 10.1080/02635143.2020.1778663

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., et al. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *Br. Med. J.* 372, n71. doi: 10.1136/bmj.n71

Pernaa, J. (2022). Possibilities and challenges of using educational cheminformatics for STEM education: a SWOT analysis of a molecular visualization engineering project. *J. Chem. Educ.* 99, 1190–1200. doi: 10.1021/acs.jchemed.1c00683

Perry, A. M. (2023). "Mutually beneficial school-industry partnerships in STEM education: an aspirational case study," in *Advances in Early Childhood and K-12 Education*, eds. J. L. Spott, L. J. Sobehrad, and R. L. Hite (Hershey, PA: IGI Global), 232–252. doi: 10.4018/978-1-6684-7771-7.ch011

Pinger, C. W., Geiger, M. K., and Spence, D. M. (2020). Applications of 3D-printing for improving chemistry education. *J. Chem. Educ.* 97, 112–117. doi: 10.1021/acs.jchemed.9b00588

Potkonjak, V., Gardner, M., Callaghan, V., Mattila, P., Guetl, C., Petrović, V. M., et al. (2016). Virtual laboratories for education in science, technology, and engineering: a review. *Comput. Educ.* 95, 309–327. doi: 10.1016/j.compedu.2016.02.002

Prishchenko, O., Cheremskaya, N., Chernogor, T., and Bukhkalo, S. (2022). Innovative methods of teaching the discipline higher mathematics to students

studying chemical technology and engineering. *ІДНРС*, 1, 30–37. doi: 10.20998/2220-4784.2022.01.05

Radianti, J., Majchrzak, T. A., Fromm, J., and Wohlgenannt, I. (2020). A systematic review of immersive virtual reality applications for higher education: design elements, lessons learned, and research agenda. *Comput. Educ.* 147, 103778. doi: 10.1016/j.compedu.2019.103778

Rahmadani, N. F., Ariani, S. R. D., Mulyani, S., and Indriyanti, N. Y. (2021). *Chemistry Teachers' Perspectives on Virtual STEM Laboratories as Learning Media*, (Yogyakarta, Indonesia: Atlantis Press). doi: 10.2991/assehr.k.210326.091

Rahmawati, Y., Zulhipri, Z., Hartanto, O., Falani, I., and Iriyadi, D. (2022). Students' conceptual understanding in chemistry learning using PhET interactive simulations. *J. Technol. Sci. Educ.* 12, 303. doi: 10.3926/jotse.1597

Raman, Y., Surif, J., and Ibrahim, N. H. (2024). The effect of problem based learning approach in enhancing problem solving skills in chemistry education: a systematic review. *Int. J. Interact. Mob. Technol.* 18, 91–111. doi: 10.3991/ijim.v18i05.47929

Rasmuson, A. (2014). *Mathematical Modeling in Chemical Engineering*. United Kingdom; New York: Cambridge University Press.

Ratamun, M. M., and Osman, K. (2018). The effectiveness of virtual lab compared to physical lab in the mastery of science process skills for chemistry experimenT. *PEC.* 76, 544–560. doi: 10.33225/pec/18.76.544

Reginald, G. (2023). Teaching and learning using virtual labs: investigating the effects on students' self-regulation. *Cogent Educ.* 10, 2172308. doi: 10.1080/2331186X.2023.2172308

Reid, N. (2008). A scientific approach to the teaching of chemistry. What do we know about how students learn in the sciences, and how can we make our teaching match this to maximise performance? *Chem. Educ. Res. Pract.* 9, 51–59. doi: 10.1039/B801297K

Rice, R. G., Do, D. D., and Maneval, J. E. (2023). *Applied Mathematics and Modeling for Chemical Engineers*. 3 edn. Hoboken, New Jersey: Wiley.

Richard, T., and Rajakumari, R. (2023). "Virtual laboratories in higher education enhancing chemistry and biology learning outcomes," in *2023 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*, (Chennai, India: IEEE)), 1–8. doi: 10.1109/ICSES60034.2023.10465586

Rivera, F. F., Pérez, T., Castañeda, L. F., and Nava, J. L. (2021). Mathematical modeling and simulation of electrochemical reactors: a critical review. *Chem. Eng. Sci.* 239, 116622. doi: 10.1016/j.ces.2021.116622

Sahin, A. (2019). "The role of interdisciplinary project-based learning in integrated STEM education," in *STEM Education 2.0*, eds. A. Sahin, and M. J. Mohr-Schroeder (Leiden: BRILL), 93–103. doi: 10.1163/9789004405400_006

Salame, I. I., and Makki, J. (2021). Examining the use of PhET simulations on students' attitudes and learning in general chemistry II. *Int. J. Env. Sci. Ed.* 17, e2247. doi: 10.21601/ijese/10966

Schuchardt, A. M., and Schunn, C. D. (2016). Modeling scientific processes with mathematics equations enhances student qualitative conceptual understanding and quantitative problem solving. *Sci. Educ.* 100, 290–320. doi: 10.1002/sce.21198

Serovajsky, S. (2021). *Mathematical Modelling*. 1st edn. Boca Raton: Chapman and Hall/CRC. doi: 10.1201/9781003035602

Serrano-Perez, J. J., González-García, L., Flacco, N., Taberner-Cortés, A., García-Arnandis, I., Pérez-López, G., et al. (2023). Traditional vs. virtual laboratories in health sciences education. *J. Biol. Educ.* 57, 36–50. doi: 10.1080/00219266.2021.1877776

Situmorang, M., Sinaga, M., Sitorus, M., and Sudrajat, A. (2022). Implementation of project-based learning innovation to develop students' critical thinking skills as a strategy to achieve analytical chemistry competencies. *IJPER.* 56, s41–s51. doi: 10.5530/ijper.56.1s.41

Smetana, L. K., and Bell, R. L. (2012). Computer simulations to support science instruction and learning: a critical review of the literature. *Int. J. Sci. Educ.* 34, 1337–1370. doi: 10.1080/09500693.2011.605182

Snyder, H. (2019). Literature review as a research methodology: an overview and guidelines. *J. Bus. Res.* 104, 333–339. doi: 10.1016/j.jbusres.2019.07.039

Staszak, M. (2023). Artificial intelligence in the modeling of chemical reactions kinetics. *Phys. Sci. Rev.* 8, 51–72. doi: 10.1515/psr-2020-0079

Sutaphan, S., and Yuenyong, C. (2019). STEM education teaching approach: inquiry from the context based. *J. Phys.: Conf. Ser.* 1340, 012003. doi: 10.1088/1742-6596/1340/1/012003

Taber, K. S. (2013). Revisiting the chemistry triplet: drawing upon the nature of chemical knowledge and the psychology of learning to inform chemistry education. *Chem. Educ. Res. Pract.* 14, 156–168. doi: 10.1039/C3RP00012E

Tan, A.-L., Ong, Y. S., Ng, Y. S., and Tan, J. H. J. (2023). STEM problem solving: inquiry, concepts, and reasoning. *Sci. Educ.* 32, 381–397. doi: 10.1007/s11191-021-00310-2

Tantayanon, S., Faikhamta, C., Prasoplarb, T., and Panyanukit, P. (2024). Teachers' perceptions and design of small-scale chemistry driven STEM learning activities. *Chem. Teach. Int.* 7, 303–317. doi: 10.1515/cti-2024-0091

Tatli, Z., and Ayas, A. (2010). Virtual laboratory applications in chemistry education. *Procedia Soc. Behav. Sci.* 9, 938–942. doi: 10.1016/j.sbspro.2010.12.263

Teixeira-Dias, J. J. C. (2017). *Molecular Physical Chemistry: A Computer-Based Approach Using Mathematica® and Gaussian*. Cham: Springer. doi: 10.1007/978-3-319-41093-7

Tytler, R. (2020). "STEM education for the twenty-first century," in *Integrated Approaches to STEM Education*, eds. J. Anderson, and Y. Li (Cham: Springer International Publishing), 21–43. doi: 10.1007/978-3-030-52229-2_3

Uludüz, ŞM, and Çalik, M. (2022). A thematic review of STEM education for disadvantaged students. *Can. J. Sci. Math. Techn. Educ.* 22, 938–958. doi: 10.1007/s42330-022-00247-w

Van Den Beemt, A., Groothuijsen, S., Ozkan, L., and Hendrix, W. (2023). Remote labs in higher engineering education: engaging students with active learning pedagogy. *J. Comput. High. Educ.* 35, 320–340. doi: 10.1007/s12528-022-09331-4

Wibawa, B., and Kardipah, S. (2018). The flipped-blended model for STEM education to improve students' performances. *IJET.* 7, 1006. doi: 10.14419/ijet.v7i2.29.14298

Wijayanti, W., Sasongko, M. N., and Kusumastuti, R. (2021). Modelling analysis of pyrolysis process with thermal effects by using Comsol Multiphysics. *Case Stud. Therm. Eng.* 28, 101625. doi: 10.1016/j.csite.2021.101625

Wilson, Z. S., McGuire, S. Y., Limbach, P. A., Doyle, M. P., Marzilli, L. G., and Warner, I. M. (2014). Diversifying science, technology, engineering, and mathematics (STEM): an inquiry into successful approaches in chemistry. *J. Chem. Educ.* 91, 1860–1866. doi: 10.1021/ed400702v

Wisudawati, A. W. (2018). Science technology engineering and mathematics (STEM) education approach against a microscopic representation skill in atom and molecule concept. *Int. J. Chem. Educ. Res.* 2 (1), 1–5. doi: 10.20885/ijcer.vol2.iss1.art1

Xie, Y., Fang, M., and Shauman, K. (2015). STEM education. *Annu. Rev. Sociol.* 41, 331–357. doi: 10.1146/annurev-soc-071312-145659

Zacharia, Z. C., Manoli, C., Xenofontos, N., De Jong, T., Pedaste, M., Van Riesen, S. A. N., et al. (2015). Identifying potential types of guidance for supporting student inquiry when using virtual and remote labs in science: a literature review. *Educ. Tech. Res. Dev.* 63, 257–302. doi: 10.1007/s11423-015-9370-0

Гулай, О, Шемет, В, Мороз, І, Саварин, П, and Кабак, В (2024). Використання ВІРТУАЛЬНОЇ ЛАБОРАТОРІЇ LABSTER ДЛЯ ВИВЧЕННЯ ХІМІЇ В ТЕХНІЧНОМУ УНІВЕРСИТЕТІ. *ITLT.* 102, 95–107. doi: 10.33407/itlt.v102i4.5737

Космодемьянская, СС, and Емельянова, АА (2025). Применение STEM и STEAM в методике химического образования: application of STEM and STEAM in chemistry education methodology. *Sci. Pract. Educ.* 6, 29–35. doi: 10.54158/27132838_2025_6_1_29